Exp: 11 Examine Network Address Translation
11/04/26 (NAT) using CISCO PACKET TRACER.

AIM: To implement Address Translation (NAT) using Cisco Packet Tracer. We'll configure static, dynamic, and Port Address Translation on a Router to enable communication between a Private Internal network and a public External network.

PROCEDURE:

Step: 1 Setup the Network Topology

1. Open Cisco Packet Tracer

2. Add devices:
   - 2 PCs (PC1 and PC2) representing inside private network
   - 1

3. Connect devices.

- Launch the Cisco Packet tracer.

Step 2: Add Devices.

Step: 3 Connect Devices

Step: 4 (optional) Verify Connections

4.1 (code)

configure inside and outside Interfaces for NAT

configure terminal Interface

gigabit Ethernet 0/0 ip nat

inside exit interface

gigabit Ethernet 0/1 ip nat

inside Exit interface

gigabit Ethernet 0/1 ip nat

outside Exit

4.2 Static Nat Configure (Example) ip

nat inside source static 192.168.1.10.200.1.1.10

4.3 Dynamic NAT configuration

for multiple inside hosts to share

multiple public IPs

- Define a NAT POOL.

ip Nat Pool MY_POOL 200.1.1.20 200.1.1.25

5. Close the Window

Repeat the exact same steps for PC2, but the IP Address stays as 192.168.1.11 and default Gateway is also 192.168.1.1

5.1 ~~6.1~~ Check NAT Translations on the router CLI:

Show IP nat translations.

You should see translations of Private IPs to Public IPs

6.2 Test Connectivity:

(From PC1 or PC2, Ping an IP on the outside network (eg: 200.1.1.2)

Result:

The router successfully translated Private IPs to Public ones using Static, dynamic, and PAT configurations.